

6 Fault List

6.1 General information about available fault codes (DTC)

These fault codes are transmitted by the ECU via parameter groups DM1 (see section 2.15) and DM2 (see section 2.16).

Each fault code is categorized in accordance with its severity and effects. This is visualized by a display. The various categories are distinguished by the alarm parameters. The project engineer can configure a fault as a red or yellow alarm. Only red and yellow alarms can be set with the parameter and only the lamp states RSL (Red Stop Lamp) and AWL (Amber Warning Lamp) are supported accordingly.

Note that the SPNs ranging between 520192 and 524287 are proprietary, the other SPNs are standard.

The FMIs (Failure Mode Indicators) have the following meaning:

- 0
Data Valid but Above Normal Operational Range – Most Severe Level
- 1
Data Valid but Below Normal Operational Range – Most Severe Level
- 11
Root Cause not known
- 15
Data Valid but Above Normal Operational Range – Least Severe Level
- 16
Data Valid but Above Normal Operational Range – Moderately Severe Level
- 17
Data Valid but Below Normal Operational Range – Least Severe Level
- 31
Condition Exists

6.2 List of available ECU fault codes (Part 1)

The table below lists all fault codes which can be acquired by the ECUs. Fault codes marked with an “x” in the columns entitled “ECU 8” and “ECU 9” are transmitted on the SAE J1939 interface of the ECU indicated.

Note that this information applies to the latest FSW release (see cover page). Some of these fault codes were not transmitted in some versions of the FSW. Refer to the columns ECU for details.

SPN	FMI	Alarm. No.	Designation	ECU 8												ECU 9 Diesel				ECU 9 Gas				ECU 9 Bifuel
				2.807.11.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3	2.917.1.2.4	2.917.1.2.5	2.917.1.2.6	2.917.1.2.7	2.907.38.1.0	
51	11	494	SD Feedback Throttle A															x	x	x	x	x		
51	31	559	AL mixture throttle A fault															x	x	x	x	x		
51	15	649	AI Req Angle Throttle A															x	x	x	x	x		
52	15	9	HI T-Coolant Intercooler		x	x	x	x	x	x	x	x	x	x	x	x	x							x
52	0	10	SS T-Coolant Intercooler		x	x	x	x	x	x	x	x	x	x	x	x	x							x
52	11	205	SD T-Coolant Intercooler		x	x	x	x	x	x	x	x	x	x	x	x	x							x
94	17	65	LO P-Fuel		x	x	x	x	x	x	x	x	x	x	x	x	x							x
94	1	66	SS P-Fuel		x	x	x	x	x	x	x	x	x	x	x	x	x							x
94	11	228	SD P-Fuel before Filter		x	x	x	x	x	x	x	x	x	x	x	x	x							x
95	15	33	HI P-Diff-Fuel		x	x	x	x	x	x	x	x	x	x	x	x	x							x
95	0	34	SS P-Diff-Fuel		x	x	x	x	x	x	x	x	x	x	x	x	x							x
95	11	239	SD P-Diff Fuel		x	x	x	x	x	x	x	x	x	x	x	x	x							x
97	15	414	HI Level Water Fuel Prefilter					x	x	x	x	x	x	x	x	x	x							x
97	11	417	SD Level Water Fuel Prefilter					x	x	x	x	x	x	x	x	x	x							x
98	15	46	LO Level Lube Oil																					
98	11	204	SD Level Lube Oil					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
98	11	636	SD Level Lube Oil J1939					x	x	x	x	x	x	x	x	x	x							x
98	17	690	AI Lube Oil Level Low											x	x	x	x							
98	17	1007	AL L1 Level Lube Oil J1939					x	x	x	x	x	x	x	x	x	x							x
98	1	1008	AL L2 Level Lube Oil J1939					x	x	x	x	x	x	x	x	x	x							x
98	0	1089	AL Oil Niveau Too High					x	x	x	x	x	x	x	x	x	x							x
98	17	1090	AL Oil Niveau Too Low					x	x	x	x	x	x	x	x	x	x							x
98	1	1091	AL Oil Niveau Far Too Low					x	x	x	x	x	x	x	x	x	x							x
98	31	1726	AL Oil Level not evaluable										x	x	x	x	x							
98	0	1727	AL L3 Oil Level High										x	x	x	x	x							
98	11	1740	AL SD_LevLubeOilSwitch										x	x	x	x								
99	15	25	HI P-Diff-Lube Oil		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
99	0	26	SS P-Diff-Lube Oil		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
99	11	221	SD P-Diff Lube Oil		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

SPN	FMI	Alarm. No.	Designation	ECU 8												ECU 9 Diesel												ECU 9 Gas												ECU 9 Bifuel																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
100	17	15	LO P-Lube Oil	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

[illegible]

SPN	FMI	Alarm. No.	Designation	ECU 8	ECU 9 Diesel										ECU 9 Gas				ECU 9 Bifuel	
					2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7			
172	11	219	SD T-Intake Air		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
172	17	457	LO T-Intake Air													x	x	x	x	x
172	1	458	LOLO T-Intake Air													x	x	x	x	x
173	11	769	SD-T-Exh. after Engine					x	x	x	x	x	x	x	x					x
173	17	800	AL L1 T-Exh. after Engine					x	x	x	x	x	x	x	x					x
173	0	1096	AL L2 T-Exh. after Engine						x	x	x	x	x	x	x					x
174	15	3	HI T-Fuel		x	x	x	x	x	x	x	x	x	x	x					x
174	0	4	SS T-Fuel		x	x	x	x	x	x	x	x	x	x	x					x
174	11	202	SD T-Fuel		x	x	x	x	x	x	x	x	x	x	x					x
174	11	694	SD T-Gas													x	x	x	x	x
174	17	719	AL T-Gas L 1													x	x	x	x	x
174	1	720	AL T-Gas L2													x	x	x	x	x
174	15	1174	HI T-Fuel f(T0)						x	x	x	x	x	x	x					
174	0	1175	SS T-Fuel f(T0)						x	x	x	x	x	x	x					
174/ 6998	15	759	AL L1 T-Fuel b.Engine					x	x	x	x	x	x	x	x					x
174/ 6998	0	760	AL L2 T-Fuel b.Engine						x	x	x	x	x	x	x					x
174/ 6998	11	761	SD T-Fuel b.Engine					x	x	x	x	x	x	x	x					x
175	15	51	HI T-Lube Oil		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
175	0	52	SS T-Lube Oil		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
175	11	216	SD T-Lube Oil		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
176	11	889	SD-T-Lubeoil ETC					x	x	x	x	x	x	x	x					x
176	0	890	AL L2 T-Lubeoil ETC					x	x	x	x	x	x	x	x					x
176	15	918	AL L1 T-Lubeoil ETC					x	x	x	x	x	x	x	x					x
188	1	90	SS Idle Speed Not Reached	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
190	0	30	SS Engine Overspeed	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
190	1	89	SS Engine Speed too Low	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
190	31	1155	SFB Emergency Stop						x	x	x	x	x	x	x					x
190	0	1156	SFB Overspeed Emergency Stop						x	x	x	x	x	x	x					x
190	31	1160	SFB Overspeed Test Emer- gency Stop						x	x	x	x	x	x	x					x
247	31	104	AL Eng Hours Counter Defect	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
250	31	102	AL Fuel Cons. Counter Defect	x	x	x	x	x	x	x	x	x	x	x	x					x
441	15	428	AL L1 T-Aux 1					x	x	x	x	x	x	x	x	x	x	x	x	x
442	15	432	AL L1 T-Aux 2																	

SPN	FMI	Alarm. No.	Designation	ECU 8										ECU 9 Diesel										ECU 9 Gas										ECU 9 Bifuel
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0											
442	0	433	AL L2 T-Aux2																															
623	31	479	AL Comb. Alarm Red (Plant)					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
624	31	478	AL Comb. Alarm Yel (Plant)					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
898	11	266	SD Speed Demand	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
898	31	836	AL Speed Demand Failure									x	x	x	x	x	x	x	x															x
966	31	596	AL Develop PR Set					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1083	15	420	AL L1 Aux 1					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1083	0	421	AL L2 Aux1					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1083	11	469	SD AUX 1					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1084	15	424	AL L1 Aux 2																															
1084	0	425	AL L2 Aux2																															
1084	11	463	SD AUX 2																															
1131	11	422	SD T-Charge Air B					x	x	x	x	x	x	x	x	x	x	x																x
1131	15	796	AL HI T-Charge Air B					x	x	x	x	x	x	x	x	x	x	x																x
1131	0	797	AL HIHI T-Charge Air B					x	x	x	x	x	x	x	x	x	x	x																x
1136	15	122	HI T-ECU	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1136	11	470	SD T-ECU	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1168	31	525	SD P-Lube Oil (R2)																															
1168	31	856	SD-P-Lubeoil ETC B					x	x	x	x	x	x	x	x	x	x																	x
1168	17	858	AL L1 P-Lubeoil ETC B					x	x	x	x	x	x	x	x	x	x																	x
1168	1	869	AL L2 P-Lubeoil ETC B					x	x	x	x	x	x	x	x	x	x																	x
1169	15	36	HI ETC2 Overspeed		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1169	16	37	SS ETC2 Overspeed	FMI 0	FMI 0	FMI 0	FMI 0	FMI 0	FMI 0	FMI 0	FMI 0	x	x	x	x	x	FMI 0	FMI 0	FMI 0	FMI 0	FMI 0	FMI 0												x
1169	11	233	SD Charger 2 Speed		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
1169	0	1729	ETC2 Overspeed L3										x	x	x	x																		
1170	15	488	HI ETC3 Overspeed		x	x	x	x	x	x	x	x	x	x	x	x	x																	x
1170	16	489	SS ETC3 Overspeed		x	x	x	x	x	x	x	x	x	x	x	x	x																	x
1170	11	806	SD Charger 3 Speed					x	x	x	x	x	x	x	x	x	x																	x
1170	0	1730	ETC3 Overspeed L3										x	x	x	x																		
1171	15	490	HI ETC4 Overspeed					x	x	x	x	x	x	x	x	x	x																	x
1171	16	491	SS ETC4 Overspeed					x	x	x	x	x	x	x	x	x	x																	x
1171	11	807	SD Charger 4 Speed					x	x	x	x	x	x	x	x	x	x																	x
1171	0	1731	ETC4 Overspeed L3										x	x	x	x																		

[illegible]

SPN	FMI	Alarm. No.	Designation	ECU 8		ECU 9 Diesel										ECU 9 Gas		ECU 9 Bifuel										
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8		2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0				
1353	0	1279	AL Knock module hard knock A2																									x
1354	11	1268	AL Knock module knock sens fail A3																									x
1354	0	1280	AL Knock module hard knock A3																									x
1355	11	1269	AL Knock module knock sens fail A4																									x
1355	0	1281	AL Knock module hard knock A4																									x
1356	11	1270	AL Knock module knock sens fail A5																									x
1356	0	1282	AL Knock module hard knock A5																									x
1357	11	1271	AL Knock module knock sens fail A6																									x
1357	0	1283	AL Knock module hard knock A6																									x
1361	11	1272	AL Knock module knock sens fail B1																									x
1361	0	1284	AL Knock module hard knock B1																									x
1362	11	1273	AL Knock module knock sens fail B2																									x
1362	0	1285	AL Knock module hard knock B2																									x
1363	11	1274	AL Knock module knock sens fail B3																									x
1363	0	1286	AL Knock module hard knock B3																									x
1364	11	1275	AL Knock module knock sens fail B4																									x
1364	0	1287	AL Knock module hard knock B4																									x
1365	11	1276	AL Knock module knock sens fail B5																									x
1365	0	1288	AL Knock module hard knock B5																									x
1366	11	1277	AL Knock module knock sens fail B6																									x

SPN	FMI	Alarm. No.	Designation	ECU 8		ECU 9 Diesel										ECU 9 Gas			ECU 9 Bifuel														
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3		2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0									
1366	0	1289	AL Knock module hard knock B6																													x	
1380	11	772	SD-Level Oil Refill Tank					x	x	x	x	x	x	x	x	x	x															x	
1380	17	805	LO Oil Level Refill					x	x	x	x	x	x	x	x	x	x															x	
1385	0	467	AL L2 T-Aux1					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x											x
1385	11	468	SD T-AUX 1					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x											x
1386	11	466	SD T-AUX 2					x	x	x	x	x	x	x	x	x	x																x
1387	17	440	AL L1 P-Aux 1					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x											x
1387	1	442	AL L2 P-Aux1					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x											x
1387	11	464	SD P-AUX 1					x	x	x	x	x	x	x	x	x	x																x
1388	17	436	AL L1 P-Aux 2																														
1388	1	437	AL L2 P-Aux2																														
1388	11	465	SD P-AUX 2																														
1390	31	1567	AL ASSV gas pressure too high																														x
1413	15	1032	AL Inj. Drift Limit 1 Cyl. A1							x	x	x	x	x	x	x	x																
1413	0	1052	AL Inj. Drift Limit 2 Cyl. A1							x	x	x	x	x	x	x	x																
1414	15	1033	AL Inj. Drift Limit 1 Cyl. A2							x	x	x	x	x	x	x	x																
1414	0	1053	AL Inj. Drift Limit 2 Cyl. A2							x	x	x	x	x	x	x	x																
1415	15	1034	AL Inj. Drift Limit 1 Cyl. A3							x	x	x	x	x	x	x	x																
1415	0	1054	AL Inj. Drift Limit 2 Cyl. A3							x	x	x	x	x	x	x	x																
1416	15	1035	AL Inj. Drift Limit 1 Cyl. A4							x	x	x	x	x	x	x	x																
1416	0	1055	AL Inj. Drift Limit 2 Cyl. A4							x	x	x	x	x	x	x	x																
1417	15	1036	AL Inj. Drift Limit 1 Cyl. A5							x	x	x	x	x	x	x	x																
1417	0	1056	AL Inj. Drift Limit 2 Cyl. A5							x	x	x	x	x	x	x	x																
1418	15	1037	AL Inj. Drift Limit 1 Cyl. A6							x	x	x	x	x	x	x	x																
1418	0	1057	AL Inj. Drift Limit 2 Cyl. A6							x	x	x	x	x	x	x	x																
1419	15	1038	AL Inj. Drift Limit 1 Cyl. A7							x	x	x	x	x	x	x	x																
1419	0	1058	AL Inj. Drift Limit 2 Cyl. A7							x	x	x	x	x	x	x	x																
1420	15	1039	AL Inj. Drift Limit 1 Cyl. A8							x	x	x	x	x	x	x	x																
1420	0	1059	AL Inj. Drift Limit 2 Cyl. A8							x	x	x	x	x	x	x	x																
1421	15	1040	AL Inj. Drift Limit 1 Cyl. A9							x	x	x	x	x	x	x	x																
1421	0	1060	AL Inj. Drift Limit 2 Cyl. A9							x	x	x	x	x	x	x	x																
1422	15	1041	AL Inj. Drift Limit 1 Cyl. A10							x	x	x	x	x	x	x	x																
1422	0	1061	AL Inj. Drift Limit 2 Cyl. A10							x	x	x	x	x	x	x	x																

SPN	FMI	Alarm. No.	Designation	ECU 8		ECU 9 Diesel										ECU 9 Gas			ECU 9 Bifuel			
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3		2.917.12.4	2.917.12.5	2.917.12.6
2602	17	1005	AL L1 Level Hydraulic Oil					x	x	x	x	x	x	x	x	x						x
2602	1	1006	AL L2 Level Hydraulic Oil					x	x	x	x	x	x	x	x	x						x
2602	31	1185	SD Level Hydraulic Oil J1939						x	x	x	x	x	x	x	x						
2631	11	517	SD P-Charge Mix before Throttle														x	x	x	x	x	
3031	11	753	SD T-RM Tank					x	x	x	x	x	x	x	x	x						x
3031	11	1134	SD T-DEF at PDM										x	x	x	x						
3031	0	1541	HI T-DEF at PDM										x	x	x	x						
3031	1	1733	LO T-DEF at PDM										x	x	x	x						
3226	17	621	LO Nox Value														x	x	x	x	x	
3226	1	622	LOLO Nox Value														x	x	x	x	x	
3226	15	623	HI Nox Value														x	x	x	x	x	
3226	0	624	HIHI Nox Value														x	x	x	x	x	
3226	11	640	SD Smart NOx Oxidation Fak-tor O2	x	x	x	x	x	x	x												x
3226	31	871	AL NOx ATO1 Communication Lost					x	x	x	x	x	x	x	x	x						x
3242	11	724	SD T-Exhaust before DPF A					x	x	x	x	x										x
3242	15	731	AL L1 T-Exhaust Before DPF					x	x	x	x	x										
3242	0	732	AL L2 T-Exhaust Before DPF					x	x	x	x	x										
3246	11	725	SD T-Exhaust after DPF A					x	x	x	x	x	x	x	x	x						x
3246	15	733	AL L1 T-Exhaust After DPF					x	x	x	x	x	x	x	x	x						
3246	0	734	AL L2 T-Exhaust After DPF					x	x	x	x	x	x	x	x	x						
3251	11	726	SD P-DeltaExhaust DPF A					x	x	x	x	x	x	x	x	x						x
3251	15	735	AL L1 P-DPF Difference					x	x	x	x	x	x	x	x	x						
3251	0	736	AL L2 P-DPF Difference					x	x	x	x	x	x	x	x	x						
3285	15	1572	AL L1 P-Diff Exhaust DOC-DPF 2										x	x	x	x						
3285	0	1573	AL L2 P-Diff Exhaust DOC-DPF 2										x	x	x	x						
3468	11	771	SD-T-Fuel B					x	x	x	x	x	x	x	x	x						x
3468	15	803	HI T-Fuel B					x	x	x	x	x	x	x	x	x						x
3468	0	804	SS T-Fuel B					x	x	x	x	x	x	x	x	x						x
3516	31	881	AL Urea Quality Release																			
3516	11	1351	SD DEF Quality Sensor										x	x	x	x						

SPN	FMI	Alarm. No.	Designation	ECU 8												ECU 9 Diesel				ECU 9 Gas				ECU 9 Bifuel
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0	
3516	31	1588	AL DEF Quality Stand. Check failed											x	x	x	x	x						
3517	11	1000	SD Level DEF Tank B						x	x	x	x												x
3517	11	1464	SD Level DEF Tank 2										x	x	x	x	x							
3543	31	891	AL turning activated															x	x	x	x	x		
3562	11	497	SD P-Charge Mix B															x	x	x	x	x		
3562	0	512	HIHI P-Charge Mix B															x	x	x	x	x		
3563	11	208	SD P-Charge Air	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
3563	15	448	HI P-Charge Air	x	x	x	x	x	x	x	x	x	x	x	x	x	x							x
3563	0	449	SS P-Charge Air	x	x	x	x	x	x	x	x	x	x	x	x	x	x							x
3668	17	44	LO Coolant Level Intercooler	x	x	x	x	x	x	x	x	x	x	x	x	x	x							x
3668	11	223	SD Level Coolant Intercooler	x	x	x	x	x	x	x	x	x	x	x	x	x	x							x
3673	11	495	SD Feedback Throttlet B															x	x	x	x	x		
3673	31	560	AL mixture throttle B fault															x	x	x	x	x		
3673	15	650	AI Req Angle Throttle B															x	x	x	x	x		
3703	11	743	AL DPF Rigorous TM Suppression					x	x	x	x	x	x	x	x	x	x							x
4076	31	526	SD T-Coolant (R2)																					
4193	11	419	SD T-Coolant b.Engine															x	x	x	x	x		
4193	15	434	HI T-Coolant before Engine															x	x	x	x	x		
4193	0	435	SS T-Coolant before Engine															x	x	x	x	x		
4202	31	528	SD Engine Speed 3rd Sensor																					
4334	11	1342	SD P-Exhaust SCR 1 Inlet										x	x	x	x	x	x						
4337	15	1075	AL F1 T - DEF too high					x	x	x	x	x	x	x	x	x	x							x
4348	15	884	AL SCR F1 SU AdBlue Quantity					x	x	x	x	x												
4358	11	1340	SD P-DiffExhaust at SCR 1											x	x	x	x	x						
4358	15	1542	HI P-Diff Exhaust SCR 1											x	x	x	x	x						
4358	0	1543	HIHI P-Diff Exhaust SCR 1											x	x	x	x	x						
4360	11	748	SD T-Exh before SCR F1					x	x	x	x	x	x	x	x	x	x							x
4360	11	783	SD T-Exh before SCR F3					x	x	x	x	x												x
4360	15	920	AL L1 T-Exhaust Before SCR F1					x	x	x	x	x	x	x	x	x	x							
4360	0	921	AL L2 T-Exhaust Before SCR F1					x	x	x	x	x	x	x	x	x	x							

SPN	FMI	Alarm. No.	Designation	ECU 8		ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel					
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7		2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6
4360	17	935	AL F1 T-Exh before SCR too LOW				x															
4360	17	1076	AL F1 T-Exh before SCR too low					x	x	x	x											x
4363	11	750	SD T-Exh after SCR F1				x	x	x	x	x											x
4363	11	784	SD T-Exh after SCR F3				x	x	x	x	x											x
4363	15	922	AL L1 T-Exhaust After SCR F1				x	x	x	x	x	x	x	x	x							
4363	0	923	AL L2 T-Exhaust After SCR F1				x	x	x	x	x	x	x	x	x							
4363	17	938	AL F1 T-Exh after SCR too LOW				x															
4363	17	1077	AL F1 T-Exh after SCR too low					x	x	x	x											x
4363	11	1462	SD T-Exhaust SCR Outlet 1								x	x	x	x	x							
4375	31	882	AL SCR F1 SU Revolution Range				x	x	x	x	x											
4377	11	1715	SD NH3 EGAT Outlet 1								x	x	x	x	x							
4377	0	1763	AL_HI3_c_NH3_EGAT_outl_F1												x							
4387	11	1343	SD P-Exhaust SCR 2 Inlet							x	x	x	x	x	x	x						
4390	15	1086	AL F2 T - DEF too high					x	x	x	x	x	x	x	x	x						x
4401	15	885	AL SCR F2 SU AdBlue Quantity				x	x	x	x	x											
4411	15	1716	HI P-Diff Exhaust SCR 2								x	x	x	x	x							

6.3 List of available ECU fault codes (Part 2)

SPN	FMI	Alarm No.	Designation	ECU 8												ECU 9 Diesel								ECU 9 Gas				ECU 9 Bifuel																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
				2.807.11.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3	2.917.1.2.4	2.917.1.2.5	2.917.1.2.6	2.917.1.2.7																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
4411	15	1716	HI P-Diff Exhaust SCR 2														x	x	x	x	x																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel										ECU 9 Gas										ECU 9 Bifuel
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.11.2.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0											
5422	17	792	AL L1 P-Charge Air B					x	x	x	x	x	x	x	x	x	x																	x
5422	1	793	AL L2 P-Charge Air B					x	x	x	x	x	x	x	x	x	x																	x
5430	11	1192	SD P-Gas before EGR Cooler							x																								
5430	11	1368	SD P-Exhaust AGR A (P6)								x	x	x	x	x	x	x																	
5430	31	1373	AL P6 Sensor A Sooting								x	x	x	x	x	x	x																	
5431	11	1193	SD P-Gas after EGR Cooler							x																								
5571	11	773	SD-P-Fuel Return path					x	x	x	x	x	x	x	x	x	x																	x
5571	17	795	AL L1 P-Fuel Return Path					x	x	x	x	x	x	x	x	x	x																	x
5841	17	1695	AL_LO1_c_RA												x	x	x																	
5841	1	1696	AL_LO2_c_RA												x	x	x																	
6586	15	1742	AL_HI1_p_Exh_SCRsys_inl_F 1												x	x	x																	
6586	0	1743	AL_HI2_p_Exh_SCRsys_inl_F 1												x	x	x																	
6587	15	1775	AL_HI1_p_Exh_SCRsys_inl_F 2												x	x	x																	
6587	0	1776	AL_HI2_p_Exh_SCRsys_inl_F 2												x	x	x																	
8160	11	1465	SD P-Exhaust DOC-DPF Inlet 1											x	x	x	x																	
8161	11	1466	SD P-Exhaust DOC-DPF Out-let 1											x	x	x	x																	
8163	11	1467	SD P-Exhaust DOC-DPF Inlet 2											x	x	x	x																	
8164	11	1468	SD P-Exhaust DOC-DPF Out-let 2											x	x	x	x																	
8165	11	1469	SD P-Diff Exhaust DOC-DPF 2											x	x	x	x																	
520303	31	1448	AL ECU Comm Lost to Act EGR A											x	x	x	x																	
520304	31	1449	AL ECU Comm Lost to Act EGR B											x	x	x	x																	
520305	31	1450	AL ECU Comm Lost to Act Byp A											x	x	x	x																	
520306	31	1451	AL ECU Comm Lost to Act Byp B											x	x	x	x																	
520307	31	1452	AL ECU Comm Lost to Act Dispenser											x	x	x	x																	
520308	31	1453	AL ECU Comm Lost to Act In-takeair											x	x	x	x																	

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel	
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3	2.917.1.2.4	2.917.1.2.5	2.917.1.2.6	2.917.1.2.7				
520309	31	1454	AL Act EGR A Comm Lost to ECU													x	x	x	x	x	x					
520310	31	1455	AL Act EGR B Comm Lost to ECU														x	x	x	x	x	x				
520311	31	1456	AL Act Byp A Comm Lost to ECU														x	x	x	x	x	x				
520312	31	1457	AL Act Byp B Comm Lost to ECU														x	x	x	x	x	x				
520313	31	1458	AL Act Dispenser Comm Lost to ECU														x	x	x	x	x	x				
520314	31	1459	AL Act Intakeair Comm Lost to ECU														x	x	x	x	x	x				
520315	31	1507	AL Flap Intake Air B Defect														x	x	x	x	x	x				
520316	31	1508	AL Act Intake Air B Ref-Drv EngStop														x	x	x	x	x	x				
520317	31	1509	AL Act Intake B Ref-Drv Eng Running														x	x	x	x	x	x				
520318	31	1510	AL Flap Intakeair B Wear Limit 1														x	x	x	x	x	x				
520319	31	1511	AL Flap Intakeair B Wear Limit 2														x	x	x	x	x	x				
520320	31	1512	AL Flap Intakeair B linkage failure														x	x	x	x	x	x				
520321	31	1513	AL Flap Intake B restricted Limit 1														x	x	x	x	x	x				
520322	31	1514	AL Flap Intake B restricted Limit 2														x	x	x	x	x	x				
520323	31	1515	AL Flap Intakeair B blocked drive														x	x	x	x	x	x				
520324	31	1516	AL Act Intake Air B Temp Alert														x	x	x	x	x	x				
520325	31	1517	AL Act Intake Air B Voltage Supply														x	x	x	x	x	x				
520326	31	1518	AL ECU Comm Lost to Act Intakeair B														x	x	x	x	x	x				
520327	31	1519	AL Act Intakeair B Comm Lost to ECU														x	x	x	x	x	x				
520328	31	1520	AL Act Intake Air B Calibr Driv Err														x	x	x	x	x	x				
520329	31	1521	AL Act Intake B Temperature Warning														x	x	x	x	x	x				

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel	
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7				
520330	31	1522	AL Act Intake Air B Targetpo- sition												x	x	x	x	x	x						
520331	31	1544	AL Blow Over Temperature Error												x	x	x	x	x	x						
520332	31	1545	AL Pmax Polynom L1												x	x	x	x	x	x						
520333	31	1546	AL Pmax Polynom L2												x	x	x	x	x	x						
520334	31	1550	AL LSU Stop Cal Airmass												x	x	x	x	x	x						
520335	31	1551	AL LSU Stop Cal Min Max												x	x	x	x	x	x						
520336	31	1552	AL LSU Stop Cal Delta Factor												x	x	x	x	x	x						
520337	31	1553	AL Act Agr A Targetpos Blocked												x	x	x	x	x	x						
520338	31	1554	AL Act Agr B Targetpos Blocked												x	x	x	x	x	x						
520339	31	1555	AL Act Byp A Targetpos Blocked												x	x	x	x	x	x						
520340	31	1556	AL Act Byp B Targetpos Blocked												x	x	x	x	x	x						
520341	31	1557	AL Act Disp Targetpos Blocked												x	x	x	x	x	x						
520342	31	1558	AL Act Intake A Targetpos Blocked												x	x	x	x	x	x						
520343	31	1559	AL Act Intake B Targetpos Blocked												x	x	x	x	x	x						
520344	31	1560	AL MB limited available												x	x	x	x	x	x						
520406	11	775	SD-P-Schmierol aft Level Pump				x	x	x	x	x	x	x	x	x	x	x	x	x	x						x
520406	17	787	AL L1 P-OilNivPump				x	x	x	x	x	x	x	x	x	x	x	x	x	x						x
520826	31	1442	AL Emergency Stopp Active												x	x	x	x	x							
520872	31	371	AL Wiring TO 1																							
520873	31	755	AL Selected Mode not Valid																							
520874	11	756	AL No Valid Mode Switch Sig- nal																							
520875	31	836	AL Speed Demand Failure																							
520876	11	698	AL SD Stop Button																							
520877	11	700	AL SD Start Button																							
520878	11	701	AL SD Up Button																							
520879	11	702	AL SD Down Button																							

SPN	FMI	Alarm No.	Designation	ECU 8		ECU 9 Diesel										ECU 9 Gas		ECU 9 Bifuel																		
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8		2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0												
520880	11	703	AL SD Ext. Speed Demand Switch																																	
520881	11	704	AL SD Speed Demand In-crease																																	
520882	11	705	AL SD Binary Speed Limita-tion																																	
520883	11	706	AL SD Droop 2 Switch																																	
520884	11	707	AL SD Frequency Switch																																	
520885	11	708	AL SD Test Overspeed																																	
520886	11	709	AL SD Override Button																																	
520887	11	710	AL SD Alarm Reset												x	x	x	x	x	x	x	x														
520888	11	711	AL SD Cylinder Cut Out																																	
520889	11	712	AL SD Request BinOut Test																																	
520890	11	713	AL SD Ext. Engine Protection																																	
520891	11	714	AL SD Prelubrication Signal																																	
520892	11	715	AL SD Ext. Increased Idle Bin																																	
520893	11	717	AL SD Request Plant DBR																																	
520900	31	321	AL Wiring Cylinder A1	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520901	31	322	AL Wiring Cylinder A2	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520902	31	323	AL Wiring Cylinder A3	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520903	31	324	AL Wiring Cylinder A4	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520904	31	325	AL Wiring Cylinder A5	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520905	31	326	AL Wiring Cylinder A6	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520906	31	327	AL Wiring Cylinder A7		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520907	31	328	AL Wiring Cylinder A8		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520908	31	329	AL Wiring Cylinder A9					x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520909	31	330	AL Wiring Cylinder A10					x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520910	31	331	AL Wiring Cylinder B1	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520911	31	332	AL Wiring Cylinder B2	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520912	31	333	AL Wiring Cylinder B3	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520913	31	334	AL Wiring Cylinder B4	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520914	31	335	AL Wiring Cylinder B5	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520915	31	336	AL Wiring Cylinder B6	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520916	31	337	AL Wiring Cylinder B7		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520917	31	338	AL Wiring Cylinder B8		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x															x
520918	31	339	AL Wiring Cylinder B9					x	x	x	x	x	x	x	x	x	x	x	x	x	x															x

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel										ECU 9 Gas										ECU 9 Bifuel	
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.11.2.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0												
520919	31	340	AL Wiring Cylinder B10					x	x	x	x	x	x	x	x	x	x																		x
520923	0	59	SS T-Coolant L3	x			x	x	x	x	x	x	x	x	x	x	x																		x
520924	15	141	AL Power too high	x				x	x	x	x	x	x	x	x	x	x																		x
520930	31	341	AL Open Load Cylinder A1	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520931	31	342	AL Open Load Cylinder A2	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520932	31	343	AL Open Load Cylinder A3	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520933	31	344	AL Open Load Cylinder A4	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520934	31	345	AL Open Load Cylinder A5	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520935	31	346	AL Open Load Cylinder A6	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520936	31	347	AL Open Load Cylinder A7		x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520937	31	348	AL Open Load Cylinder A8		x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520938	31	349	AL Open Load Cylinder A9					x	x	x	x	x	x	x	x	x	x																		x
520939	31	350	AL Open Load Cylinder A10					x	x	x	x	x	x	x	x	x	x																		x
520940	31	351	AL Open Load Cylinder B1	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520941	31	352	AL Open Load Cylinder B2	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520942	31	353	AL Open Load Cylinder B3	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520943	31	354	AL Open Load Cylinder B4	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520944	31	355	AL Open Load Cylinder B5	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520945	31	356	AL Open Load Cylinder B6	x	x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520946	31	357	AL Open Load Cylinder B7		x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520947	31	358	AL Open Load Cylinder B8		x	x	x	x	x	x	x	x	x	x	x	x	x																		x
520948	31	359	AL Open Load Cylinder B9					x	x	x	x	x	x	x	x	x	x																		x
520949	31	360	AL Open Load Cylinder B10					x	x	x	x	x	x	x	x	x	x																		x
520952	31	381	AL Wiring TOP 1		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520953	31	382	AL Wiring TOP 2		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520954	31	383	AL Wiring TOP 3		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520955	31	384	AL Wiring TOP 4		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520958	31	400	AL Open Load Digital Input 1		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520959	31	401	AL Open Load Digital Input 2		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520960	31	402	AL Open Load Digital Input 3		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520961	31	403	AL Open Load Digital Input 4		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520962	31	404	AL Open Load Digital Input 5		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520963	31	405	AL Open Load Digital Input 6		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520964	31	406	AL Open Load Digital Input 7		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520965	31	407	AL Open Load Digital Input 8		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

SPN	FMI	Alarm No.	Designation	ECU 8	ECU 9 Diesel												ECU 9 Gas		ECU 9 Bifuel													
					2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8		2.917.1.2.3	2.917.1.2.4	2.917.1.2.5	2.917.1.2.6	2.917.1.2.7	2.907.38.1.0							
520970	31	536	AL Wiring PWM_CM1		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520971	31	558	AL Wiring PWM_CM2		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520972	31	581	AL Wiring PWM_CM3		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520973	31	587	AL Wiring PWM_CM4		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520974	31	597	AL Wiring PWM_CM5		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520975	31	608	AL Wiring PWM_CM6		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520976	31	609	AL Wiring PWM_CM7		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520977	31	626	AL Wiring PWM_CM8		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520978	31	627	AL Wiring PWM_CM9		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520979	31	628	AL Wiring PWM_CM10		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520982	17	410	LO U-PDU	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520982	1	411	LOLO U-PDU	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520982	15	412	HI U-PDU	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520982	0	413	HIHI U-PDU	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520982	11	444	SD U-PDU	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520983	31	610	AL Wiring Suction Restritor 1	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520984	31	611	AL Wiring Suction Restritor 2	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520985	31	612	AL Wiring Pressure Control Valve 1	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520986	31	613	AL Wiring Pressure Control Valve 2	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x														x
520990	31	476	AL Crash Rec. Init. Error	x				x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
520991	31	692	AL ECU Power OFF/ON Re- quired	x																												
520992	31	696	AL Smart Connect USB Error	x																												
520993	31	697	AL Smart Connect RS485 Er- ror	x																												
520994	11	763	AL OL ASO Flap Feedback B			x	x	x	x	x	x	x	x	x	x	x	x	x														x
520995	11	764	AL ASO Flap B closed, A failed			x	x	x	x	x	x	x	x	x	x	x	x	x														x
520996	31	765	AL OL ASO Flap Feedback A			x	x	x	x	x	x	x	x	x	x	x	x	x														x
520997	31	766	AL ASO Flap A closed, B failed			x	x	x	x	x	x	x	x	x	x	x	x	x														x
520998	31	767	AL ASO Flaps closed			x	x	x	x	x	x	x	x	x	x	x	x	x														x
520999	31	778	AL ASO Flaps open / failed to close			x	x	x	x	x	x	x	x	x	x	x	x	x														x

SPN	FMI	Alarm No.	Designation	ECU 8	ECU 9 Diesel											ECU 9 Gas			ECU 9 Bifuel																		
					2.807.11.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8		2.917.1.2.3	2.917.1.2.4	2.917.1.2.5	2.917.1.2.6	2.917.1.2.7	2.907.38.1.0												
521000	31	839	AL ASO Flap A Not Closed by ECU			x	x	x	x	x	x	x	x	x	x	x	x																				x
521001	31	81	AL Rail Leakage			x	x	x	x	x	x	x	x	x	x	x	x	x																			x
521002	1	91	SS Release Speed Not Reached			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521003	1	92	SS Starter Speed Not Reached			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521004	1	93	SS T-Preheat			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521004	17	94	LO T-Preheat			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521005	31	840	AL ASO Flap B Not Closed by ECU				x	x	x	x	x	x	x	x	x	x	x	x																			x
521006	31	180	AL CAN1 Node Lost			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521007	31	181	AL CAN2 Node Lost			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521008	31	182	AL CAN Wrong Parameters			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521009	31	183	AL CAN No PU-Data			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521010	31	184	AL CAN PU-Data Flash Error			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521011	31	186	AL CAN1 Bus Off			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521012	31	187	AL CAN1 Error Passive			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521013	31	188	AL CAN2 Bus Off			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521014	31	189	AL CAN2 Error Passive			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521016	31	229	AL Stop Camshaft Sensor De- fect			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521017	11	230	SD Crankshaft Speed			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521018	11	231	SD Camshaft Speed			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521019	11	270	SD Frequency Input				x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521020	31	361	AL Power Stage Low			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x																	x
521021	31	362	AL Power Stage High			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x																x
521022	31	363	AL Stop Power Stage			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x																x
521023	31	365	AL Stop MV-Wiring Ground			x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x																x
521023	15	455	AL L1 Aux1 Plant																																		x
521023	0	456	AL L2 Aux1 Plant																																		x
521024	31	408	AL Open Load Emerg. Stop Input ESI				x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521025	11	450	SD Idle/End-Torque Input [%]				x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521026	31	454	SS Power Reduction Active				x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521027	31	472	AL Stop SD				x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

[illegible]

SPN	FMI	Alarm No.	Designation	ECU 8		ECU 9 Diesel												ECU 9 Gas				ECU 9 Bifuel	
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6		2.917.12.7
521063	15	614	AL L1 P-Fuel Add.sec.fuelfilt. Diff		x	x	x	x	x	x	x	x	x	x	x	x	x						x
521063	11	625	SD P-Fuel bef. Add.sec.fuelfilter		x	x	x	x	x	x	x	x	x	x	x	x	x						x
521064	11	643	SD P-Fuel b.o.F.		x	x	x	x	x	x	x	x	x	x	x	x	x						x
521064	15	644	AL L1 P-Fuel b.o.F.		x	x	x	x	x	x	x	x	x	x	x	x	x						x
521064	0	645	AL L2 P-Fuel b.o.F.		x	x	x	x	x	x	x	x	x	x	x	x	x						x
521067	31	833	AL Emission Warning					x	x	x	x	x	x	x	x	x	x						x
521068	31	834	AL Gas Path Warning					x	x	x	x	x	x	x	x	x	x						x
521069	31	835	AL Gas Path Fault					x	x	x	x	x	x	x	x	x	x						x
521070	31	953	AL GPE Lambda value invalid					x	x	x	x	x	x	x	x	x	x						
521071	31	954	AL NOx value invalid					x	x	x	x	x	x	x	x	x	x						
521072	31	955	AL Thermal Management active					x	x	x	x	x	x	x	x	x	x						
521073	31	956	AL p5 ctrlvar lower limit active					x	x	x	x	x	x	x	x	x	x						
521074	31	957	AL p5 ctrlvar max BOI min active					x	x	x	x	x	x	x	x	x	x						
521075	31	958	AL Lambda ctrlvar limit min active					x	x	x	x	x	x	x	x	x	x						
521076	31	959	AL Lambda ctrlvar max BOI min act					x	x	x	x	x	x	x	x	x	x						
521077	31	960	AL Nox p5 min BOI max active					x	x	x	x	x	x	x	x	x	x						
521078	31	961	AL NOx p5 max BOI min active					x	x	x	x	x	x	x	x	x	x						
521080	31	962	AL GPS Lambda ctrlvar max active					x	x	x	x	x	x	x	x	x	x						
521081	31	963	AL GPS p5 ctrlvar max active					x	x	x	x	x	x	x	x	x	x						
521082	31	964	AL GPS p5 ctrlvar min active					x	x	x	x	x	x	x	x	x	x						
521083	31	855	AL Bypass Throttle 2 Defect					x	x	x	x	x	x	x	x	x	x						x
521084	31	837	AL Bypass Valve Defect					x	x	x	x	x	x	x	x	x	x						x
521085	31	718	AL Intake AirThrottle Defect					x	x	x	x	x	x	x	x	x	x						x
521086	11	754	SD Bosch LSU Lambda Sensor					x	x	x	x	x	x	x	x	x	x						x
521087	31	630	AL EGR Throttle B Defect					x	x	x	x	x	x	x	x	x	x						x
521088	17	727	AL L1 Delta T-NT Intercooler					x	x	x	x	x	x	x	x	x	x						x
521088	0	728	AL L2 Delta T-NT Intercooler					x	x	x	x	x	x	x	x	x	x						x

SPN	FMI	Alarm No.	Designation	ECU 8		ECU 9 Diesel										ECU 9 Gas		ECU 9 Bifuel																
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8		2.917.1.2.3	2.917.1.2.4	2.917.1.2.5	2.917.1.2.6	2.917.1.2.7	2.907.38.1.0										
521089	31	757	AL Lim T-Coolant LT Fan					x	x	x																								x
521091	31	39	AL ETC2 CutIn Failure					x	x	x	x	x	x	x	x	x	x																	x
521092	31	95	AL Prelubrication Fault					x	x	x	x	x	x	x	x	x	x	x	x	x	x													x
521093	31	142	AL MCR exceeded 1 hour					x	x	x	x	x	x	x	x	x	x																	x
521094	31	190	AL EMU Parameter Not Supported					x	x	x	x	x	x	x	x	x	x	x	x	x	x													x
521095	11	268	SD Spinning Value					x	x	x	x	x	x	x	x	x	x																	x
521096	31	390	AL MCR exceeded					x	x	x	x	x	x	x	x	x	x	x	x	x	x													x
521097	31	441	AL Rail 2 Leakage					x	x	x	x	x	x	x	x	x	x																	x
521098	15	460	HI T-Exhaust EMU					x	x	x	x	x	x	x	x	x	x																	x
521098	17	461	LO T-Exhaust EMU					x	x	x	x	x	x	x	x	x	x																	x
521099	15	462	HI T-Coolant EMU					x	x	x	x	x	x	x	x	x	x																	x
521100	11	471	SD Coil Current					x	x	x	x	x	x	x	x	x	x																	x
521103	31	492	AL ETC4 CutIn Failure					x	x	x	x	x	x	x	x	x	x																	x
521104	31	493	AL ETC3 CutIn Failure					x	x	x	x	x	x	x	x	x	x																	x
521105	11	500	AL Wiring POM Starter 1					x	x	x	x	x	x	x	x	x	x																	x
521106	11	501	AL Wiring POM Starter 2					x	x	x	x	x	x	x	x	x	x																	x
521107	11	502	AL Open Load POM Alternator					x	x	x	x	x	x	x	x	x	x																	x
521108	17	801	AL L1 T-Raw water after Pump					x	x	x	x	x	x	x	x	x	x																	x
521109	11	504	AL CAN POM Node Lost					x	x	x	x	x	x	x	x	x	x																	x
521110	1	506	AL Low Starter Voltage					x	x	x	x	x	x	x	x	x	x																	x
521111	31	507	AL POM Error					x	x	x	x	x	x	x	x	x	x																	x
521112	31	508	AL Wrong POM-ID					x	x	x	x	x	x	x	x	x	x																	x
521113	31	514	Write Error Flash					x	x	x	x	x	x	x	x	x	x	x	x	x	x													x
521114	31	519	Oillevel Calibration Error					x	x	x	x	x	x	x	x	x	x																	x
521115	11	520	SD P-Intake Air after Filter A					x	x	x	x	x	x	x	x	x	x	x	x	x	x													x
521116	11	522	SD P-Intake Air after Filter B					x	x	x	x	x	x	x	x	x	x	x	x	x	x													x
521117	0	551	SS Engine Overspeed Camshaft					x	x	x	x	x	x	x	x	x	x																	x
521118	11	577	SD T-Lube Oil Pan					x	x	x	x	x	x	x	x	x	x																	x
521118	17	593	AL T-Lube Oil Pan Low					x	x	x	x	x	x	x	x	x	x																	x
521119	17	586	LO P-Oil Refill Pump					x	x	x	x	x	x	x	x	x	x																	x
521119	11	588	SD P-Oil Refill Pump					x	x	x	x	x	x	x	x	x	x																	x
521120	11	600	SD T-Exhaust A+B					x	x	x	x	x	x	x	x	x	x																	x

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel										ECU 9 Gas										ECU 9 Bifuel		
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7	2.907.38.1.0													
521121	11	637	SD T-Lube Oil Pan J1939					x	x	x	x	x	x	x	x	x	x																		x	
521122	31	693	AL MB Valve error					x	x	x	x	x	x	x	x	x	x																		x	
521123	15	737	AL L1 P-DPF Norm Difference					x	x	x	x	x	x	x	x	x	x																		x	
521123	0	738	AL L2 P-DPF Norm Difference					x	x	x	x	x	x	x	x	x	x																		x	
521123	17	739	AL L3 P-DPF Norm Difference					x	x	x	x	x																							x	
521123	1	740	AL L4 P-DPF Norm Difference					x	x	x	x	x																							x	
521124	11	741	AL DPF Rigorous TM Aborted					x	x	x	x	x	x	x	x	x	x																		x	
521125	11	742	AL DPF Periodic Rigorous TM					x	x	x	x	x	x	x	x	x	x																		x	
521126	11	744	AL DPF Flash Read Error					x	x	x	x	x	x	x	x	x	x																			
521127	11	758	AL DEF Nozzle Damage					x	x	x	x	x	x	x	x	x	x																			
521128	11	762	AL Smart Connect Lost					x	x	x	x	x	x	x	x	x	x	x	x	x	x														x	
521129	11	770	SD-T-Sea water after Pump					x	x	x	x	x	x	x	x	x	x																		x	
521131	11	776	SD-P-Schmierol at HP Pump A					x	x	x	x	x	x	x	x	x	x																		x	
521132	11	777	SD-P-Schmierol at HP Pump B					x	x	x	x	x	x	x	x	x	x																		x	
521133	11	808	SD Charger 5 Speed					x	x	x	x	x	x	x	x	x	x																		x	
521134	11	809	AL F1 NOx before SCR SensorDefect					x	x	x	x	x																								x
521134	31	810	AL F1 NOx before SCR Comm Lost					x	x	x	x	x																								x
521134	11	1477	SD NOx EGAT Inlet 1											x	x	x	x	x																		
521134	31	1478	AL Comm Lost NOx EGAT Inlet 1											x	x	x	x	x																		
521135	11	811	AL F1 NOx after SCR Sensor-Defect					x	x	x	x	x																								x
521135	31	812	AL F1 NOx afterSCR Comm Lost					x	x	x	x	x																								x
521135	11	1483	SD NOx EGAT Outlet 1											x	x	x	x	x																		
521135	31	1484	AL Comm Lost NOx EGAT Outlet 1											x	x	x	x	x																		
521136	11	813	AL F2 NOx before SCR SensorDefect					x	x	x	x	x																								x
521136	31	814	AL F2 NOx before SCR Comm Lost					x	x	x	x	x																								x
521136	11	1479	SD NOx EGAT Inlet 2											x	x	x	x	x																		
521136	31	1480	AL Comm Lost NOx EGAT Inlet 2											x	x	x	x	x																		

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas		ECU 9 Bifuel				
				ECU 8										ECU 9 Diesel								ECU 9 Gas						
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3	2.917.12.4	2.917.12.5	2.917.12.6	2.917.12.7						
521137	11	815	AL F2 NOx after SCR Sensor-Defect					x	x	x	x	x																x
521137	31	816	AL F2 NOx after SCR Comm Lost						x	x	x	x	x															x
521137	11	1485	SD NOx EGAT Outlet 2													x	x	x	x	x								
521137	31	1486	AL Comm Lost NOx EGAT Outlet 2													x	x	x	x	x								
521138	11	817	AL F3 NOx before SCR SensorDefect						x	x	x	x	x															x

6.4 List of available ECU fault codes (Part 3)

SPN	FMI	Alarm No.	Designation	ECU 8	ECU 9 Diesel										ECU 9 Gas			ECU 9 Bifuel				
				2.807.11.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8		2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6
521138	31	818	AL F3 NOx before SCR Comm Lost					x	x	x	x	x										x
521138	11	1481	SD NOx EGAT Inlet 3										x	x	x	x	x					
521138	31	1482	AL Comm Lost NOx EGAT In-let 3										x	x	x	x	x					
521139	11	819	AL F3 NOx after SCR Sensor-Defect					x	x	x	x	x										x
521139	31	820	AL F3 NOx after SCR Comm Lost					x	x	x	x	x										x
521139	11	1487	SD NOx EGAT Outlet 3										x	x	x	x	x					
521139	31	1488	AL Comm Lost NOx EGAT Outlet 3										x	x	x	x	x					
521140	31	821	HI ETC1 Idle Speed too High					x	x	x	x	x	x	x	x	x	x					x
521141	31	822	HI ETC2 Idle Speed too High					x	x	x	x	x	x	x	x	x	x					x
521142	31	823	HI ETC3 Idle Speed too High					x	x	x	x	x	x	x	x	x	x					x
521143	31	824	HI ETC4 Idle Speed too High					x	x	x	x	x	x	x	x	x	x					x
521144	31	825	HI ETC5 Idle Speed too High					x	x	x	x	x	x	x	x	x	x					x
521145	31	826	AL ETC1 Speed Deviation					x	x	x	x	x	x	x	x	x	x					x
521146	31	827	AL ETC2 Speed Deviation					x	x	x	x	x	x	x	x	x	x					x
521147	31	828	AL ETC3 Speed Deviation					x	x	x	x	x	x	x	x	x	x					x
521148	31	829	AL ETC4 Speed Deviation					x	x	x	x	x	x	x	x	x	x					x
521149	31	830	AL ETC5 Speed Deviation					x	x	x	x	x	x	x	x	x	x					x
521150	31	831	AL ETC Job Rotation					x	x	x	x	x	x	x	x	x	x					x
521151	31	832	AL EIL Different Engine Num-ber					x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
521152	31	838	AL ash volume					x	x	x	x	x	x	x	x	x	x					
521153	31	843	SD T-Charge Air before EGR					x	x	x	x	x	x	x	x	x	x					x
521153	15	844	AL HI T-Charge Air before EGR					x	x	x	x	x	x	x	x	x	x					x
521153	0	845	AL HIHI T-Charge Air before EGR					x	x	x	x	x	x	x	x	x	x					x
521154	15	846	AL HI T-Charge Air Diff AB					x	x	x	x	x	x	x	x	x	x					x
521154	0	847	AL HIHI T-Charge Air Diff AB					x	x	x	x	x	x	x	x	x	x					x
521155	31	851	AL External Start and HD too high					x	x	x	x	x	x	x	x	x	x					x

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel										ECU 9 Gas										ECU 9 Bifuel
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6	2.917.1.2.3.7	2.907.38.1.0											
521156	31	852	AL Max. Blank Shot time expired					x	x	x	x	x	x	x	x	x	x																x	
521157	31	853	AL HSB1 Communication Lost					x	x	x	x	x	x	x	x	x	x																x	
521158	31	854	AL HSB1 Actuator defect					x	x	x	x	x	x	x	x	x	x																x	
521159	31	857	AL NOx ATO1 Sensor Defect					x	x	x	x	x	x	x	x	x	x																x	
521160	31	859	AL HSB2 Communication Lost					x	x	x	x	x																					x	
521161	31	860	AL HSB2 Actuator defect					x	x	x	x	x																					x	
521162	31	861	Defect in DEF pipe sns/act system					x	x	x	x	x																						
521163	31	862	DEF Tank ht. sns/act defect					x																										
521164	31	863	AL HSB3 Communication Lost					x	x	x																							x	
521165	31	864	AL HSB3 Actuator defect					x	x	x																							x	
521166	31	865	AL HSB4 Communication Lost					x	x	x																							x	
521167	31	866	AL HSB4 Actuator defect					x	x	x																							x	
521168	31	870	AL MB Valve defect 2					x	x	x	x	x	x	x	x	x	x																x	
521169	31	872	AL EGR A Reference learn failed					x	x	x																							x	
521170	31	873	AL Urea Tank Level Empty*					x	x	x	x	x																					x	
521171	31	876	AL EGR B Reference learn failed					x	x	x																							x	
521172	31	877	AL Bypass A Reference learn failed					x	x	x																							x	
521173	31	878	AL Bypass B Fast learn failed					x	x	x																							x	
521174	31	879	AL Dispenser Reference learn failed					x	x	x																							x	
521175	31	880	AL Intake Throttle Ref learn failed					x	x	x																							x	
521176	31	886	AL SCR AdBlue pressure					x	x	x	x	x																					x	
521177	31	892	AL Flow 1 Supply Unit 1 Comm Lost					x	x	x	x	x																					x	
521178	31	893	AL Flow 1 Supply Unit 2 Comm Lost					x	x	x	x	x																					x	
521179	31	894	AL Flow 2 Supply Unit 1 Comm Lost					x	x	x	x	x																					x	

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3.3	2.917.12.3.4	2.917.12.3.5	2.917.12.3.6	2.917.12.3.7			
521180	31	895	AL Flow 2 Supply Unit 2 Comm Lost					x	x	x	x	x												x	
521181	31	896	AL Flow 3 Supply Unit 1 Comm Lost					x	x	x	x	x												x	
521182	31	897	AL Flow 3 Supply Unit 2 Comm Lost					x	x	x	x	x												x	
521183	31	898	AL Trican Communication Lost					x	x	x	x	x	x	x	x	x	x							x	
521184	31	899	AL OLT Communication Lost					x	x	x	x	x	x	x	x	x	x							x	
521185	31	900	AL SCR F3 SU Revolution Range																						
521186	31	901	AL SCR F3 SU AdBlue Quantity																						
521187	15	902	HI T-Coolant Cylinder Head					x	x	x	x	x	x	x	x	x	x							x	
521187	11	903	AL SD T Coolant at cylinder head					x	x	x	x	x	x	x	x	x	x							x	
521187	16	904	SS T-Coolant Cylinder Head					x	x	x	x	x	x	x	x	x	x							x	
521188	31	905	AL AdBlue expected consume failed					x																	
521188	31	1018	AL F1 DEF consumption error					x	x	x	x	x	x	x	x	x	x							x	
521189	31	906	AL AdBlue balance failed					x																	
521189	31	1019	AL F1 DEF balance error					x	x	x	x	x	x	x	x	x	x							x	
521190	31	907	AL NOx raw gas emission failed					x																	
521190	31	1020	AL F1 Raw gas emission error					x	x	x	x	x	x	x	x	x	x							x	
521191	31	908	AL Approach NOx Dosing Stop failed					x																	
521191	31	1021	AL F1 NOx Annaeherung error					x	x	x	x	x	x	x	x	x	x							x	
521192	31	909	AL SCR T-Exh between flows failed					x																	
521192	31	1022	AL T-Exh bef SCR between F1 and F2					x	x	x	x													x	
521193	31	910	AL SCR Expected T-Exh bef failed					x																	
521193	31	1028	AL F1 Erw T-Abg vor SCR Error					x	x	x	x	x	x	x	x	x	x							x	
521194	31	911	AL SCR Expected T-Exh after failed					x																	

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas		ECU 9 Bifuel
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6	2.917.1.2.3.7		
521194	31	1072	AL F1 Exp T-Exh aft SCR error						x	x	x	x	x	x	x	x								x
521195	31	912	AL SCR F1 T-Exh bef gradient					x																
521195	31	1073	AL F1 grad T-Exh bef SCR error						x	x	x	x	x	x	x	x	x							x
521196	31	913	AL SCR F2 T-Exh bef gradient					x																
521196	31	1084	AL F2 grad T-Exh bef SCR error						x	x	x	x	x	x	x	x	x							x
521197	31	914	AL SCR F3 T-Exh bef gradient																					
521198	31	915	AL SCR F1 T-Exh after gradient					x																
521198	31	1074	AL F1 grad T-Exh aft SCR error						x	x	x	x	x	x	x	x	x							x
521199	31	916	AL SCR F2 T-Exh after gradient					x																
521199	31	1085	AL F2 grad T-Exh aft SCR error						x	x	x	x	x	x	x	x	x							x
521200	31	917	AL SCR F3 T-Exh after gradient																					
521201	15	928	AL L1 T-Exhaust Before SCR F3						x	x	x	x	x	x	x	x	x							x
521201	0	929	AL L2 T-Exhaust Before SCR F3						x	x	x	x	x	x	x	x	x							x
521201	17	937	AL F3 T-Exh before SCR too LOW																					
521202	15	930	AL L1 T-Exhaust After SCR F3						x	x	x	x	x	x	x	x	x							x
521202	0	931	AL L2 T-Exhaust After SCR F3						x	x	x	x	x	x	x	x	x							x
521202	17	940	AL F3 T-Exh after SCR too LOW																					
521203	17	941	AL SCR operating temperature tooLOW						x	x	x	x	x	x	x	x	x							x
521204	17	942	AL Cataly conversion too lowF1						x	x	x	x	x	x	x	x	x							x
521205	17	943	AL Cataly conversion too lowF2						x	x	x	x	x	x	x	x	x							x
521206	17	944	AL Cataly conversion too lowF3						x	x	x	x	x	x	x	x	x							x
521207	31	947	AL Invalid LSI Channel Config						x	x	x	x	x	x	x	x	x							x
521208	31	950	AL SCR SU fault(s) exist						x	x	x	x	x											
521209	31	951	AL ETC0 CutIn Failure						x	x	x	x	x	x	x	x	x							x

SPN	FMI	Alarm No.	Designation	ECU 8												ECU 9 Diesel								ECU 9 Gas				ECU 9 Bifuel
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6	2.917.1.2.3.7						
521235	0	572	L3 AI CAN Max. Retarded Timing															x	x	x	x	x						
521236	31	583	AL Circuit Breaker Closed															x	x	x	x	x						
521237	31	604	AL Hut Changespeed max.															x	x	x	x	x						
521238	17	617	LO Actual Value Hu															x	x	x	x	x						
521238	1	618	LOLO Actual Value Hu															x	x	x	x	x						
521238	15	619	HI Actual Value Hu															x	x	x	x	x						
521238	0	620	HIHI Actual Value Hu															x	x	x	x	x						
521239	31	646	AI Knock Intensity															x	x	x	x	x						
521240	31	651	AL Preheating Error															x	x	x	x	x						
521241	31	652	AL GET Comm Lost															x	x	x	x	x						
521242	31	653	AL IC92x Comm Lost															x	x	x	x	x						
521243	31	654	AL FSeries Comm Lost							x	x	x	x					x	x	x	x	x						
521244	31	655	AL TecJet Comm Lost															x	x	x	x	x						
521245	31	656	AL ProActA Comm Lost															x	x	x	x	x						
521246	31	657	AL ProActB Comm Lost															x	x	x	x	x						
521247	31	658	AL NOxA Comm Lost															x	x	x	x	x						
521248	31	659	AL NOxB Comm Lost															x	x	x	x	x						
521249	31	664	AL Oil Refill Error															x	x	x	x	x						
521250	31	665	AL GET Yellow															x	x	x	x	x						
521251	31	666	AL IC92x Yellow															x	x	x	x	x						
521252	31	667	AL FSeries Yellow							x	x	x	x					x	x	x	x	x						
521253	31	668	AL TecJet Yellow															x	x	x	x	x						
521254	31	669	AL ProActA Yellow															x	x	x	x	x						
521255	31	670	AL ProActB Yellow															x	x	x	x	x						
521256	31	671	AL NOxA Yellow															x	x	x	x	x						
521257	31	672	AL NOxB Yellow															x	x	x	x	x						
521258	31	675	AL GET Red															x	x	x	x	x						
521259	31	676	AL IC92x Red															x	x	x	x	x						
521260	31	677	AL FSeries Red							x	x	x	x					x	x	x	x	x						
521261	31	678	AL TecJet Red															x	x	x	x	x						
521262	31	679	AL ProActA Red															x	x	x	x	x						
521263	31	680	AL ProActB Red															x	x	x	x	x						
521264	31	681	AL NOxA Red															x	x	x	x	x						
521265	31	682	AL NOxB Red															x	x	x	x	x						

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6	2.917.1.2.3.7	2.907.38.1.0		
521353	19	985	AL NOx ATO2 Communication Lost						x	x	x	x	x	x	x	x									x
521354	17	1011	AL DEF Tank Level low**						x	x	x	x													x
521355	31	1015	AL Total breakdown NOx sensors						x	x	x	x	x	x	x	x	x								
521356	31	1016	AL Redundancy loss NOx sensors						x	x															
521357	31	1027	AL Engine Cold Active						x	x	x	x	x	x	x	x	x								x
521358	11	1092	AL Engine Coolant Temperature SD						x	x	x	x													
521359	11	1093	AL Intake Air Temperature SD						x	x	x	x													
521360	11	1094	AL DEF Tank Temperature SD						x	x	x	x													
521361	31	1095	AL Engine Coolant Valve Defect(DEF)						x	x	x	x													
521361	31	1698	AL Engine Coolant Valve Defect											x	x	x	x	x							
521362	31	1097	AL Flap Egr A Communication Lost						x	x	x	x	x	x	x	x	x								x
521363	0	1098	AL Flap Egr A Temperature too high						x	x	x	x	x	x	x	x	x								x
521364	31	1099	AL Flap Egr A Targetposition						x	x	x	x	x	x	x	x	x								x
521365	31	1100	AL Flap Egr B Communication Lost						x	x	x	x	x	x	x	x	x								x
521366	0	1101	AL Flap Egr B Temperature too high						x	x	x	x	x	x	x	x	x								x
521367	31	1102	AL Flap Egr B Targetposition						x	x	x	x	x	x	x	x	x								x
521368	31	1103	AL Flap BypassA Communication Lost						x	x	x	x	x	x	x	x	x								x
521369	0	1104	AL Flap BypassA Temperature too high						x	x	x	x	x	x	x	x	x								x
521370	31	1105	AL Flap Bypass A Targetposition						x	x	x	x	x	x	x	x	x								x
521371	31	1106	AL Flap BypassB Communication Lost						x	x	x	x	x	x	x	x	x								x
521372	0	1107	AL Flap BypassB Temperature too high						x	x	x	x	x	x	x	x	x								x
521373	31	1108	AL Flap Bypass B Targetposition						x	x	x	x	x	x	x	x	x								x
521374	31	1109	AL Flap Dispens Communication Lost						x	x	x	x	x	x	x	x	x								x

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel									
				2.807.11.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6	2.917.1.2.3.7	2.907.38.1.0											
521375	0	1110	AL Flap DispensTemperature too high								x	x	x	x	x	x	x	x																x
521376	31	1111	AL Flap Dispenser Targetposi- tion								x	x	x	x	x	x	x	x	x															x
521377	31	1112	AL Flap Intake Communica- tion Lost								x	x	x	x	x	x	x	x	x															x
521378	0	1113	AL Flap Intake Temperature too high								x	x	x	x	x	x	x	x	x															x
521379	31	1114	AL Flap Intake Air Targetposi- tion								x	x	x	x	x	x	x	x	x															x
521380	31	1120	AL Flap Egr A Calibration Drive Err								x	x	x	x	x	x	x	x	x															x
521381	31	1121	AL Flap Egr B Calibration Drive Err								x	x	x	x	x	x	x	x	x															x
521382	31	1122	AL Flap BypassA Calibr. Drive Err								x	x	x	x	x	x	x	x	x															x
521383	31	1123	AL Flap BypassB Calibr. Drive Err								x	x	x	x	x	x	x	x	x															x
521384	31	1124	AL Flap Dispenser Calibr Drive Err								x	x	x	x	x	x	x	x	x															x
521385	31	1125	AL Flap Intake Air Calibr Drive Err								x	x	x	x	x	x	x	x	x															x
521386	15	1126	AL L1 PCV Defect								x	x	x	x	x	x	x	x	x															x
521386	0	1127	AL L2 PCV Defect								x	x	x	x	x	x	x	x	x															x
521387	15	1128	AL L1 PCV2 Defect								x	x	x	x	x	x	x	x	x															x
521387	0	1129	AL L2 PCV2 Defect								x	x	x	x	x	x	x	x	x															x
521388	6	1130	AL Short Circuit Analog Out 1								x	x	x	x	x	x	x	x	x															x
521389	6	1131	AL Short Circuit Analog Out 2								x	x	x	x	x	x	x	x	x															x
521426	15	1167	AL Inplausible NOx High Lim- it1								x	x	x	x	x	x	x	x	x															
521426	17	1168	AL Inplausible NOx Low Lim- it1								x	x	x	x	x	x	x	x	x															
521426	0	1172	AL Inplausible NOx High Lim- it2								x	x	x	x	x	x	x	x	x															
521426	1	1173	AL Inplausible NOx Low Lim- it2								x	x	x	x	x	x	x	x	x															
521427	6	1139	AL Communication Error ECU - EXU								x	x	x	x	x	x	x	x	x															
521428	31	1029	AL IAP Missing Energization Data								x	x	x	x	x	x	x	x	x															

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas		ECU 9 Bifuel									
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6	2.917.1.2.3.7											
521429	31	1026	AL IAP Communication Lost									x	x	x	x	x	x	x															
521432	31	29	HI ETC Idle Speed too High									x	x	x	x	x	x	x															
521433	31	38	AL ETC Speed Deviation									x	x	x	x	x	x	x															
521434	15	61	HI P-AdCrank Case L1									x	x	x	x	x	x	x															
521434	0	62	HI P-AdCrank Case L2									x	x	x	x	x	x	x															
521435	15	85	HI T-Umbblasen									x	x	x	x	x	x	x															
521435	0	86	SS T-Umbblasen									x	x	x	x	x	x	x															
521435	11	241	SD T-Umbblasen									x	x	x	x	x	x	x															
521436	11	495	SD Feedback Throttlet A									x	x	x	x	x	x	x															
521437	31	983	AL Stop on Trigger Crashre-corder									x	x	x	x	x	x	x															
521438	11	1136	SD P-DEF Filter Out at PDM													x	x	x	x														
521438	17	1140	AL PDM DEF Pressure low									x	x	x	x	x	x	x															
521438	15	1141	AL PDM DEF Pressure high									x	x	x	x	x	x	x															
521439	31	1142	SFB Relay Interface Defect									x	x	x	x	x	x	x															x
521440	31	1143	SFB Relay Voltage Level Low									x	x	x	x	x	x	x															x
521441	31	1144	SFB SpeedSensorRedundancyLost									x	x	x	x	x	x	x															x
521442	31	1145	SFB Speed Sensor defect									x	x	x	x	x	x	x															x
521443	31	1146	SFB Flap Feedback Interface A									x	x	x	x	x	x	x															x
521444	31	1147	SFB Flap Feedback Interface B									x	x	x	x	x	x	x															x
521445	31	1148	SFB ESS_H Interface defect									x	x	x	x	x	x	x															x
521446	31	1149	SFB SFS Interface defect									x	x	x	x	x	x	x															x
521447	31	1150	SFB ECU_CAN Communication lost									x	x	x	x	x	x	x															x
521448	31	1151	SFB SFB_CAN_CommunicationLost									x	x	x	x	x	x	x															x
521449	31	1152	SFB Configuration Data failure									x	x	x	x	x	x	x															x
521450	31	1153	SFB_Power Supply Out of Range									x	x	x	x	x	x	x															x
521451	31	1154	SFB SFB_Defect									x	x	x	x	x	x	x															x
521452	31	1158	SFB Flap A failed to close									x	x	x	x	x	x	x															x
521453	31	1159	SFB Flap B failed to close									x	x	x	x	x	x	x															x
521454	31	1161	SFB Flap A not closed by SFB									x	x	x	x	x	x	x															

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3.3	2.917.12.3.4	2.917.12.3.5	2.917.12.3.6	2.917.12.3.7			
521455	31	1162	SFB Flap B not closed by SFB							x	x	x	x	x	x	x							x		
521456	31	1163	SFB Flap A closed							x	x	x	x	x	x	x							x		
521457	31	1164	SFB Flap B closed							x	x	x	x	x	x	x							x		
521458	31	1165	SFB SFS_Not available							x	x	x	x	x	x	x							x		
521459	31	1166	SFB SFB/ECU_Engine Speed deviation							x	x	x	x	x	x	x							x		
521461	31	1170	AL Permanent Injection							x	x	x	x	x	x	x									
521462	31	1171	AL Permanent Injection 2							x	x	x	x	x	x	x									
521464	31	1178	AL J1939 Heartbeat1 Lost							x	x	x	x	x	x	x							x		
521465	31	1179	AL J1939 Heartbeat2 Lost							x	x	x	x	x	x	x							x		
521466	31	1180	AL NOx 2 Sensor invalid							x	x	x	x	x	x	x									
521467	15	1181	Alarm NOx Deviation of flow 1/2 L1							x	x	x	x	x	x	x									
521467	0	1182	Alarm NOx Deviation of flow 1/2 L2							x	x	x	x	x	x	x									
521468	31	1183	AL Invalid Measurement Config							x	x	x	x	x	x	x							x		
521469	31	1184	AL HLT Sensor Communication Lost							x	x	x	x	x	x	x									
521470	0	1186	AL EGR FlpA Wear Limit2							x															
521471	0	1187	AL EGR FlpB Wear Limit2							x															
521472	0	1188	AL Bypass FlpA Wear Limit2							x															
521473	0	1189	AL Bypass FlpB Wear Limit2							x															
521474	0	1190	AL Dispenser Flp Wear Limit2							x															
521475	0	1191	AL IntakeAir Flp Wear Limit2							x															
521476	31	1201	AL Stop Start Low Power Supply							x	x	x	x	x	x	x									
521480	31	1194	AL Deviation red. ECU P-Exhaust																				x		
521482	31	1253	AL TecJet A defect																				x		
521483	31	1254	AL TecJet B defect																				x		
521484	31	1255	AL gas supply																				x		
521485	31	1256	AL TecJet A Com lost																				x		
521486	31	1257	AL TecJet B Com lost																				x		
521487	31	1258	AL TecJet soiled																				x		
521488	31	1259	AL ProAct A defect																				x		
521489	31	1260	AL ProAct B defect																						

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6	2.917.1.2.3.7			
521490	31	1261	AL ProAct charge air press implau																					x	
521491	31	1262	AL ProAct A com lost																					x	
521492	31	1263	AL ProAct B com lost																					x	
521493	31	1264	AL Knock module com lost																					x	
521494	31	1265	AL Knock module speed sensor fail																					x	
521495	31	1290	AL Knock module defect																					x	
521496	31	1291	AL EMU9 Com Lost																					x	
521499	31	1353	AL BiFuel Gas Quality																					x	
521506	31	1426	AL EGR T-Intake-Air T-Coolant On							x	x	x	x	x	x	x	x								
521507	31	1427	AL EGR T-Intake-Air T-Fuel On							x	x	x	x	x	x	x	x								
521508	31	1428	AL EGR T-Intake T-Coolant Reduction							x	x	x	x	x	x	x	x								
521509	31	1429	AL EGR T-Intake T-Fuel Reduction							x	x	x	x	x	x	x	x								
521510	31	1430	AL Act Agr A Temperature Alert							x	x	x	x	x	x	x	x								
521511	31	1431	AL Act Agr B Temperature Alert							x	x	x	x	x	x	x	x								
521512	31	1432	AL Act Byp A Temperature Alert							x	x	x	x	x	x	x	x								
521513	31	1433	AL Act Byp B Temperature Alert							x	x	x	x	x	x	x	x								
521514	31	1434	AL Act Dispenser Temperature Alert							x	x	x	x	x	x	x	x								
521515	31	1435	AL Act Intake Air Temperature Alert							x	x	x	x	x	x	x	x								
521516	31	1436	AL Act Agr A Voltage Supply							x	x	x	x	x	x	x	x								
521517	31	1437	AL Act Agr B Voltage Supply							x	x	x	x	x	x	x	x								
521518	31	1438	AL Act Byp A Voltage Supply							x	x	x	x	x	x	x	x								
521519	31	1439	AL Act Byp B Voltage Supply							x	x	x	x	x	x	x	x								
521520	31	1440	AL Act Dispenser Voltage Supply							x	x	x	x	x	x	x	x								
521521	31	1441	AL Act Intake Air Voltage Supply							x	x	x	x	x	x	x	x								
521522	31	1376	AL Refdrive Impossibl Ubat Low							x	x	x	x	x	x	x	x								

SPN	FMI	Alarm No.	Designation	ECU 8		ECU 9 Diesel										ECU 9 Gas		ECU 9 Bifuel																	
				2.807.11.1.3	2.907.11.1.5	2.907.11.1.6	2.907.11.1.7	2.907.11.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8		2.917.12.3.3	2.917.12.3.4	2.917.12.3.5	2.917.12.3.6	2.917.12.3.7	2.907.38.1.0											
521523	31	1377	AL Refdrive Impossibl Eng Op Point												x	x	x	x	x	x															
521524	31	1378	AL Act Egr A Ref-Drv EngStop												x	x	x	x	x	x															
521525	31	1379	AL Act Egr B Ref-Drv EngStop												x	x	x	x	x	x															
521526	31	1380	AL Act Byp A Ref-Drv EngStop												x	x	x	x	x	x															
521527	31	1381	AL Act Byp B Ref-Drv EngStop												x	x	x	x	x	x															
521528	31	1382	AL Act Dispenser Ref-Drv EngStop												x	x	x	x	x	x															
521529	31	1383	AL Act Intake Air Ref-Drv Eng-Stop												x	x	x	x	x	x															
521530	31	1384	AL Act Egr A Ref-Drv Eng Running												x	x	x	x	x	x															
521531	31	1385	AL Act Egr B Ref-Drv Eng Running												x	x	x	x	x	x															
521532	31	1386	AL Act Byp A Ref-Drv Eng Running												x	x	x	x	x	x															
521533	31	1387	AL Act Byp B Ref-Drv Eng Running												x	x	x	x	x	x															
521534	31	1388	AL Act Disp B Ref-Drv Eng Running												x	x	x	x	x	x															
521535	31	1389	AL Act Intake Ref-Drv Eng Running												x	x	x	x	x	x															
521536	31	1390	AL Flap Egr A Wear Limit 1												x	x	x	x	x	x															
521537	31	1391	AL Flap Egr B Wear Limit 1												x	x	x	x	x	x															
521538	31	1392	AL Flap Byp A Wear Limit 1												x	x	x	x	x	x															
521539	31	1393	AL Flap Byp B Wear Limit 1												x	x	x	x	x	x															
521540	31	1394	AL Flap Disp Wear Limit 1												x	x	x	x	x	x															
521541	31	1395	AL Flap Intakeair Wear Limit 1												x	x	x	x	x	x															
521542	31	1396	AL Flap Egr A Wear Limit 2												x	x	x	x	x	x															
521543	31	1397	AL Flap Egr B Wear Limit 2												x	x	x	x	x	x															
521544	31	1398	AL Flap Byp A Wear Limit 2												x	x	x	x	x	x															
521545	31	1399	AL Flap Byp B Wear Limit 2												x	x	x	x	x	x															
521546	31	1400	AL Flap Disp Wear Limit 2												x	x	x	x	x	x															
521547	31	1401	AL Flap Intakeair Wear Limit 2												x	x	x	x	x	x															
521548	31	1402	AL Flap Egr A linkage failure												x	x	x	x	x	x															
521549	31	1403	AL Flap Egr B linkage failure												x	x	x	x	x	x															

SPN	FMI	Alarm No.	Designation	ECU 8		ECU 9 Diesel										ECU 9 Gas		ECU 9 Bifuel			
				2.807.1.1.3	2.907.1.1.5	2.907.1.1.6	2.907.1.1.7	2.907.1.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8		2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5
521550	31	1404	AL Flap Byp A linkage failure							x	x	x	x	x	x	x					
521551	31	1405	AL Flap Byp B linkage failure							x	x	x	x	x	x	x					
521552	31	1406	AL Flap Dispenser linkage failure							x	x	x	x	x	x	x					

6.5 List of available ECU fault codes (Part 4)

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel
				2.807.11.1.3	2.907.38.1.5	2.907.38.1.6	2.907.38.1.7	2.907.38.1.8	2.907.1.2.0	2.907.1.2.1	2.907.1.2.2	2.907.1.2.3	2.907.1.2.4	2.907.1.2.5	2.907.1.2.6	2.907.1.2.7	2.907.1.2.8	2.917.1.2.3.3	2.917.1.2.3.4	2.917.1.2.3.5	2.917.1.2.3.6	2.917.1.2.3.7			
521670	31	1315	AL SCR1 Hose3 Defect											x	x	x	x	x							
521671	31	1316	AL SCR1 Hose4 Defect											x	x	x	x	x							
521672	31	1317	AL SCR1 Hose5 Defect											x	x	x	x	x							
521673	31	1324	AL SCR2 Hose2 Defect											x	x	x	x	x							
521674	31	1325	AL SCR3 Hose2 Defect											x	x	x	x	x							
521675	31	1326	AL SCR2 Hose3 Defect											x	x	x	x	x							
521676	31	1327	AL SCR3 Hose3 Defect											x	x	x	x	x							
521677	31	1328	AL SCR2 Hose4 Defect											x	x	x	x	x							
521678	11	1333	SD T-Exhaust before DOC 3											x	x	x	x	x							
521679	11	1344	SD T-AmbientAir at SCR 1											x	x	x	x	x							
521680	11	1345	SD T-AmbientAir at SCR 2											x	x	x	x	x							
521681	11	1346	SD T-AmbientAir at SCR 3											x	x	x	x	x							
521682	11	1347	SD T-AmbientAir at DEFTank											x	x	x	x	x							
521683	11	1348	SD Level Leakage DEF Tank											x	x	x	x								
521684	31	1306	AL Hilite PDM Piston 1 Defect											x	x	x	x	x							
521685	31	1309	AL PDM DEF Hose Flow De- fect											x	x	x	x	x							
521686	31	1310	AL PDM DEF Hose Return De- fect											x	x	x	x	x							
521687	31	1349	AL DEF Tank Hose Flow De- fect											x	x	x	x	x							
521688	31	1350	AL DEF Tank Hose Return De- fect											x	x	x	x	x							
521689	31	1352	AL DEF Tank External Heating Defect											x	x	x	x	x							
521690	31	1444	AL EGR cooler sooting A											x	x	x	x	x							
521691	31	1445	AL EGR flap A											x	x	x	x	x							
521704	11	1460	SD T-Exhaust SCR Inlet 3											x	x	x	x	x							
521705	11	1463	SD T-Exhaust SCR Outlet 3											x	x	x	x	x							
521706	11	1470	SD P-Exhaust DOC-DPF Inlet 3											x	x	x	x	x							
521707	11	1471	SD P-Diff Exhaust DOC-DPF 3											x	x	x	x	x							
521708	11	1472	SD P-Exhaust DOC-DPF Out- let 3											x	x	x	x	x							
521709	15	1473	AL L1 T-Exhaust DOC Inlet 2											x	x	x	x	x							

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel					ECU 9 Gas				
				2.807.11.1.3	2.907.38.1.5	2.907.38.1.6	2.907.38.1.7	2.907.38.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3.3	2.917.12.3.4	2.917.12.3.5	2.917.12.3.6	2.917.12.3.7	2.907.38.1.0 ECU 9 Bifuel
521709	0	1474	AL L2 T-Exhaust DOC Inlet 2											x	x	x	x	x					
521710	15	1475	AL L1 T-Exhaust DOC Inlet 3											x	x	x	x	x					
521710	0	1476	AL L2 T-Exhaust DOC Inlet 3											x	x	x	x	x					
521711	31	1491	AL Defect HeatCartridge DEF Doser 1											x	x	x	x	x					
521712	31	1494	AL Defect HeatCartridge DEF Doser 2											x	x	x	x	x					
521713	31	1497	AL Defect HeatCartridge DEF Doser 3											x	x	x	x	x					
521714	31	1500	AL Defect HeatCartridge DEF Doser 4											x	x	x	x	x					
521715	31	1503	AL Defect HeatCartridge DEF Doser 5											x	x	x	x	x					
521716	31	1506	AL Defect HeatCartridge DEF Doser 6											x	x	x	x	x					
521733	31	1528	AL DEF Tank Heating Cart Flow Defec											x	x	x	x	x					
521734	31	1529	AL DEFTank Heat.Cart. Return Defect											x	x	x	x	x					
521735	31	1530	AL F3 Unexpect DEF consumption											x	x	x	x	x					
521736	31	1531	AL F3 DEF balance out of Range											x	x	x	x	x					
521737	31	1532	AL F3 Unexpect NOx Inlet value											x	x	x	x	x					
521738	31	1533	AL F3 No Nox Approach Inl-Outl											x	x	x	x	x					
521739	31	1534	AL F3 T - DEF too high											x	x	x	x	x					
521740	31	1535	AL F3 Unexp. SCR T-Exh. Inlet											x	x	x	x	x					
521752	15	1561	AL EPL: ISO ICxN Power Time Limit1											x	x	x	x	x					
521752	0	1562	AL EPL: ISO ICxN Power Time Limit2											x	x	x	x	x					
521753	31	1563	AL EDP: Max. Power reached											x	x	x	x	x					
521754	31	1564	AL EDP: Max. Speed reached											x	x	x	x	x					
521755	31	1565	AL EDP: Max. Torque reached											x	x	x	x	x					
521756	31	1566	AL Leakage Level DEF Tank											x	x	x	x	x					
521757	31	1568	AL SCR Emerg Override Reset Require											x	x	x	x	x					

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel
				2.807.11.1.3	2.907.38.1.5	2.907.38.1.6	2.907.38.1.7	2.907.38.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3.3	2.917.12.3.4	2.917.12.3.5	2.917.12.3.6	2.917.12.3.7			
521758	31	1569	AL Max Overheating Counts Exceeded												x	x	x	x	x						
521759	31	1570	AL DEF Hose Tank Feed 2.1 Defect												x	x	x	x	x						
521760	31	1590	AL DEF Hose Tank Return 2.1 Defect												x	x	x	x	x						
521761	15	1591	HI T-Charge Air Compressor Out												x	x	x	x	x						
521762	31	1592	AL J1939 AddressClaiming Conflict												x	x	x	x	x						
521763	31	1598	AL EGAT Bypass MainFlow-Flap stopped												x	x	x	x	x						
521764	31	1599	AL EGAT BypFlp Wear Limit 1												x	x	x	x	x						
521765	31	1600	AL EGAT BypFlp Wear Limit 2												x	x	x	x	x						
521766	31	1601	AL EGAT BypFlp linkage fail-ure												x	x	x	x	x						
521767	31	1602	AL EGAT BypFlp restricted Limit 1												x	x	x	x	x						
521768	31	1603	AL EGAT BypFlp restricted Limit 2												x	x	x	x	x						
521769	31	1604	AL EGAT BypFlp blocked drive												x	x	x	x	x						
521770	31	1605	AL EGAT BypFlp Temperature Warning												x	x	x	x	x						
521771	31	1606	AL EGAT BypFlp Temperature Alert												x	x	x	x	x						
521772	31	1607	AL EGAT BypFlp Hallsensor Error												x	x	x	x	x						
521773	31	1608	AL EGAT BypFlp SW Init Error												x	x	x	x	x						
521774	31	1609	AL EGAT BypFlp Power Sup- ply Warning												x	x	x	x	x						
521775	31	1610	AL EGAT BypFlp Power Sup- ply Alert												x	x	x	x	x						
521776	31	1611	AL EGAT BypFlp Target Pos Deviation												x	x	x	x	x						
521777	31	1612	AL EGAT BypFlp Target Pos blocked												x	x	x	x	x						
521778	31	1613	AL EGAT BypFlp Comm Loss to Flap												x	x	x	x	x						
521779	31	1614	AL EGAT BypFlp Reference drv fail												x	x	x	x	x						

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel					ECU 9 Gas				
				2.807.11.1.3	2.907.38.1.5	2.907.38.1.6	2.907.38.1.7	2.907.38.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3.3	2.917.12.3.4	2.917.12.3.5	2.917.12.3.6	2.917.12.3.7	2.907.38.1.0 ECU 9 Bifuel
521780	31	1615	AL EGAT BypFlp Calibration drv fail										x	x	x	x	x						
521781	31	1700	AL SU Defect DEF Doser 1										x	x	x	x	x						
521782	31	1701	AL EGAT BypFlp Comm Lost to ECU										x	x	x	x	x						
521783	31	1702	AL AGN BypKlp Configuration failed										x	x	x	x	x						
521784	31	1703	AL Hydraulics Fault DEF Doser 1										x	x	x	x	x						
521785	31	1704	AL SU Defect DEF Doser 2										x	x	x	x	x						
521786	31	1705	AL Hydraulics Fault DEF Doser 2										x	x	x	x	x						
521787	31	1706	AL SU Defect DEF Doser 3										x	x	x	x	x						
521788	31	1707	AL Hydraulics Fault DEF Doser 3										x	x	x	x	x						
521789	31	1708	AL SU Defect DEF Doser 4										x	x	x	x	x						
521790	31	1709	AL Hydraulics Fault DEF Doser 4										x	x	x	x	x						
521791	31	1710	AL SU Defect DEF Doser 5										x	x	x	x	x						
521792	31	1711	AL Hydraulics Fault DEF Doser 5										x	x	x	x	x						
521793	31	1712	AL SU Defect DEF Doser 6										x	x	x	x	x						
521794	31	1713	AL Hydraulics Fault DEF Doser 6										x	x	x	x	x						
521795	31	1714	AL CommLost NH3 EGAT Outlet 1										x	x	x	x	x						
521796	15	1718	HI P-Diff Exhaust SCR 3										x	x	x	x	x						
521796	0	1719	HIHI P-Diff Exhaust SCR 3										x	x	x	x	x						
521798	0	1576	HI mt-DEF Actual DEF Doser										x	x	x	x	x						
521798	17	1577	LO mt-DEF Actual DEF Doser										x	x	x	x	x						
521798	1	1578	LOLO mt-DEF Actual DEF Doser										x	x	x	x	x						
521799	31	1589	AL DEF Quality noStand Check failed										x	x	x	x	x						
521800	31	1195	AL PDM LeakCheck Failed										x	x	x	x	x						
521801	31	1527	AL PDM Pump Blocked										x	x	x	x	x						
521802	31	1536	AL F3 Unexp. SCR T-Exh. Outlet										x	x	x	x	x						

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel								ECU 9 Gas			ECU 9 Bifuel
				2.807.11.1.3	2.907.38.1.5	2.907.38.1.6	2.907.38.1.7	2.907.38.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3.3	2.917.12.3.4	2.917.12.3.5	2.917.12.3.6	2.917.12.3.7			
521803	31	1537	AL F3 SCR T-Exh Inl gradient hi									x	x	x	x	x									
521804	31	1538	AL F3 SCRT-Exh Outl gradient hi									x	x	x	x	x									
521805	11	1734	SD p_RawWater									x	x	x	x										
521805	17	1736	AL p_RawWater L1									x	x	x	x										
521805	1	1737	AL p_RawWater L2									x	x	x	x										
521806	31	1738	AL P5 Or P6 Implausible									x	x	x	x										
521807	31	1739	AL Comm Lost EMU9									x	x	x	x										
521841	31	1202	AL P-Exhaust in IMO2-Mode																				x		
521846	15	1756	AL_HI1_t_EngineLow-Load_SCRsys									x	x	x	x										
521846	16	1757	AL_HI2_t_EngineLow-Load_SCRsys									x	x	x	x										
521846	0	1758	AL_HI3_t_EngineLow-Load_SCRsys									x	x	x	x										
521847	15	1759	AL_HI4_t_EngineLow-Load_SCRsys									x	x	x	x										
521848	15	1760	AL_HI1_t_EngineLowLoad									x	x	x	x										
521859	15	1761	AL_HI1_load_HC_DPFsys									x	x	x	x										
521859	0	1762	AL_HI2_load_HC_DPFsys									x	x	x	x										
521860	0	992	AL HIHI P-Charge Mix Mean to electr																		x	x			
521860	1	993	AL LOLO P-Charge Mix Mean to electr																		x	x			
521941	15	1750	AL_HI1_p_ChargeAir_Deviation									x	x	x											
521942	15	1751	AL_HI1_load_HC_short_SCR sys									x	x	x											
521942	0	1752	AL_HI2_load_HC_short_SCR sys									x	x	x											
521944	31	994	AL ENGINE TORSIONAL VIBRATION									x	x	x											
521945	31	1771	AL_CommLost_combi_sensor_RA_tank									x	x	x											
521945	11	1772	AL_SD_combi_sensor_RA_tank									x	x	x											
521946	11	1777	AL_SD_EBCO_Resistor_Plant									x	x	x											
521948	11	987	SD-T-Raw water after Engine												x										

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel				ECU 9 Gas		ECU 9 Bifuel
				2.807.11.1.3	2.907.38.1.5	2.907.38.1.6	2.907.38.1.7	2.907.38.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3.3	2.917.12.3.4	
521948	15	988	AL L1 T-Raw water after Engine														x			
521948	0	1200	AL L2 T-Raw water after Engine														x			
521949	31	1169	AL NOx Alarm Fail Counter														x			
521950	31	1525	SD P-Exhaust SCR Outlet 2														x			
521951	15	1753	AL_HI1_load_HC_long_SCRsys														x			
521952	15	1754	AL_HI1_load_RA_SCRsys														x			
521952	16	1755	AL_HI2_load_RA_SCRsys														x			
521953	31	1773	AL Security Mode PCV Open														x			
521954	0	1778	AL Inplausible NOx High Limit3														x			
521955	31	1783	AL_Comm-Lost_NH3_EGAT_outl_F2														x			
521956	31	1788	AL Security Mode PCV Open 2														x			
521957	15	1789	AL_HI1_B_InplausibleNOxF2														x			
521957	16	1790	AL_HI2_B_InplausibleNOxF2														x			
521957	0	1791	AL_HI3_B_InplausibleNOxF2														x			
521958	15	1792	AL_HI1_B_InplausibleNOxF3														x			
521958	16	1793	AL_HI2_B_InplausibleNOxF3														x			
521958	0	1794	AL_HI3_B_InplausibleNOxF3														x			
521959	15	1795	AL_HI_N_EventTimeMissing-Counter_F2														x			
521960	15	1796	AL_HI_N_EventTimeMissing-Counter_F3														x			
521961	15	1720	AL_HI1_Per_Rig_TM_DPFsys_F2														x			
521962	15	1721	AL_HI1_Per_Rig_TM_DPFsys_F3														x			
t.b.d.		995	HI P-Coolant before Engine***																x	
t.b.d.		996	L4 AI CAN Max. Retarded Timing FG***																x	
t.b.d.		997	AL FastGas Max Activation Time***																x	

SPN	FMI	Alarm No.	Designation	ECU 8										ECU 9 Diesel					ECU 9 Gas			ECU 9 Bifuel
				2.807.11.1.3	2.907.38.1.5	2.907.38.1.6	2.907.38.1.7	2.907.38.1.8	2.907.12.0	2.907.12.1	2.907.12.2	2.907.12.3	2.907.12.4	2.907.12.5	2.907.12.6	2.907.12.7	2.907.12.8	2.917.12.3.3	2.917.12.3.4	2.917.12.3.5	2.917.12.3.6	
t.b.d.		998	AL LO p LubeOil after Filt. FastGas***																			x
t.b.d.		999	AL HI T Exhaust mean Fast-Gas***																			x

* IMPORTANT

According the regulation 97/68/EG (Release 2011), Appendix I "8 Requirements for the type approval Stage III B and IV" a warning has to indicate, when the tank level goes under 1 % of the tank capacity.

** IMPORTANT

According the regulations of 97/68/EG (Release 2011), Appendix I "8 Requirements for the type approval Stage III B and IV" a warning has to indicate, when the tank level goes under 10 % of the tank capacity.

*** IMPORTANT

SW Version 2.7 actually for field-trial engines (06/2019)